LAB 4: Activity Diagram

Smart Waste Collection Management System

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1. Objective

To model the dynamic behaviour of the Smart Waste Collection Management System as UML activity diagrams, showing the flow of activities, decisions and parallel actions between the actors of the system, based on SRS v1.0.

2. Activity diagrams included

Figure	Scenario	SRS reference
Diagram 1 (page 2)	Bin fill-level monitoring, threshold alert, route optimization and collection by the driver.	4.1, 4.2, 4.3, 5.5 (REQ-BM-1 to 3, REQ-AL-1 to 3, REQ-RO-1 to 3)
Diagram 2 (page 3)	Citizen complaint: submission, tracking ID, admin action, collection and automatic resolution.	4.5, 5.5 (REQ-CF-1 to 3)

3. Swimlanes (actors)

Swimlane	Responsibility in the diagrams
Sensor Node (IoT)	Measures distance, converts it to fill percentage, retries failed reads and transmits data every 15 minutes.
Backend Server	Stores readings, checks the threshold, generates and logs alerts, computes routes, assigns complaint tracking IDs and resets bins.
Administrator	Views alerts, requests or edits routes and reviews citizen complaints on the dashboard.
Driver (Mobile App)	Receives the route, collects waste and marks each bin as collected.
Citizen	Submits a complaint with location and optional photo and views its status.

4. Key decision points

- **Reading valid?** - if the sensor read fails, it is retried and logged (REQ-BM-3).
- **Fill-level at or above threshold?** - only then an alert is generated and the bin becomes eligible for a route (REQ-AL-2, SRS 5.5).
- **Adjust route?** - the administrator may add or remove bins from the generated route (REQ-RO-3).
- **More bins on route?** - the driver repeats collection until all bins on the route are done.
- **Fork and join (Diagram 2)** - after logging, the citizen gets the tracking ID while the administrator acts on the complaint in parallel; both paths join when the bin is collected and the complaint is auto-resolved.

5. Notation used

Symbol	Meaning
Filled circle	Initial node (start of the activity)
Rounded rectangle	Action / activity
Diamond	Decision node with guard labels (Yes / No)
Thick horizontal bar	Fork (split into parallel flows) or join (synchronise)
Circle with filled centre	Activity final node (end)
Arrow / vertical columns	Control flow / swimlanes (actor responsible for each action)

Activity Diagram 1 - Bin Monitoring, Alerting and Waste Collection

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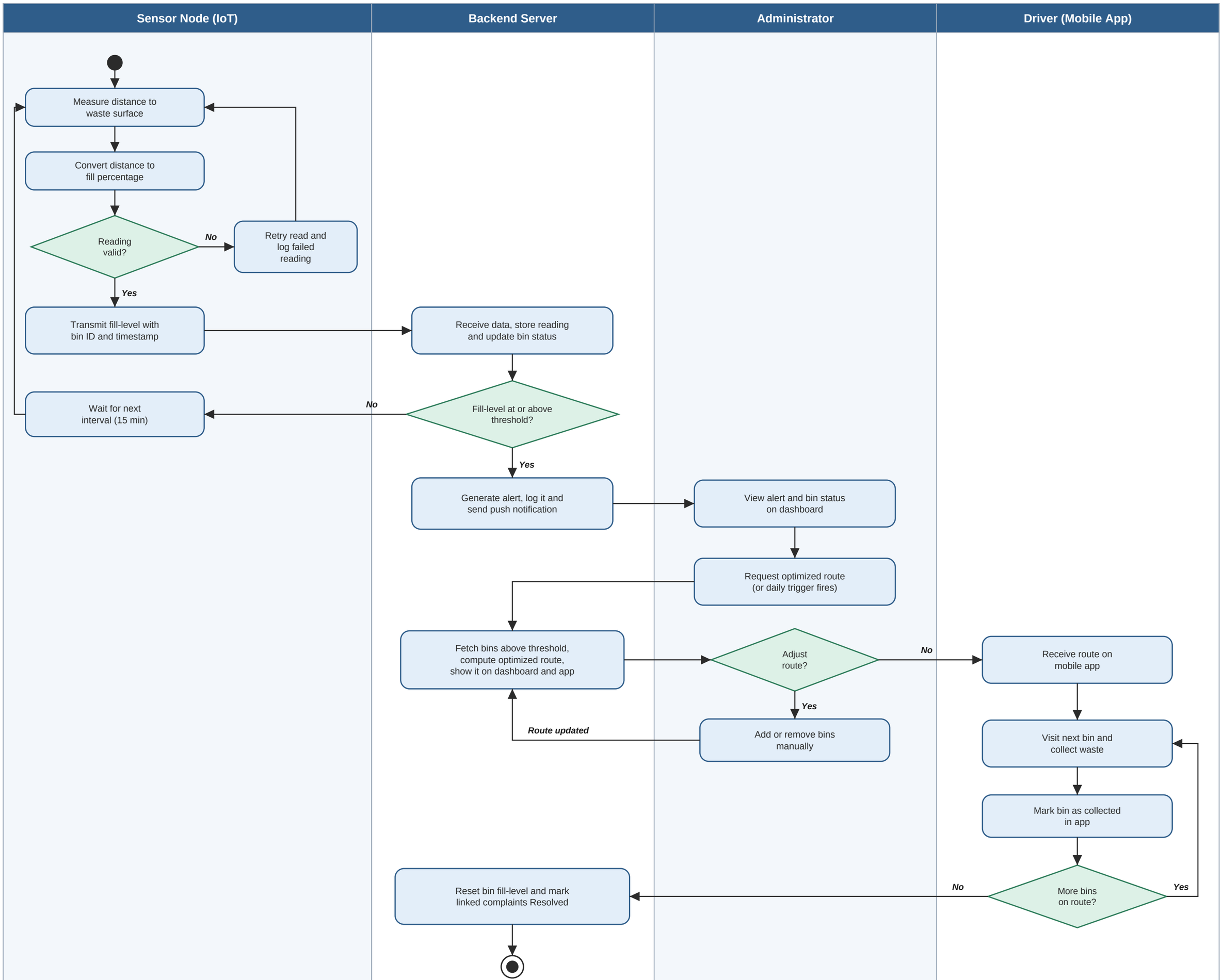

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Activity Diagram 2 - Citizen Complaint Handling

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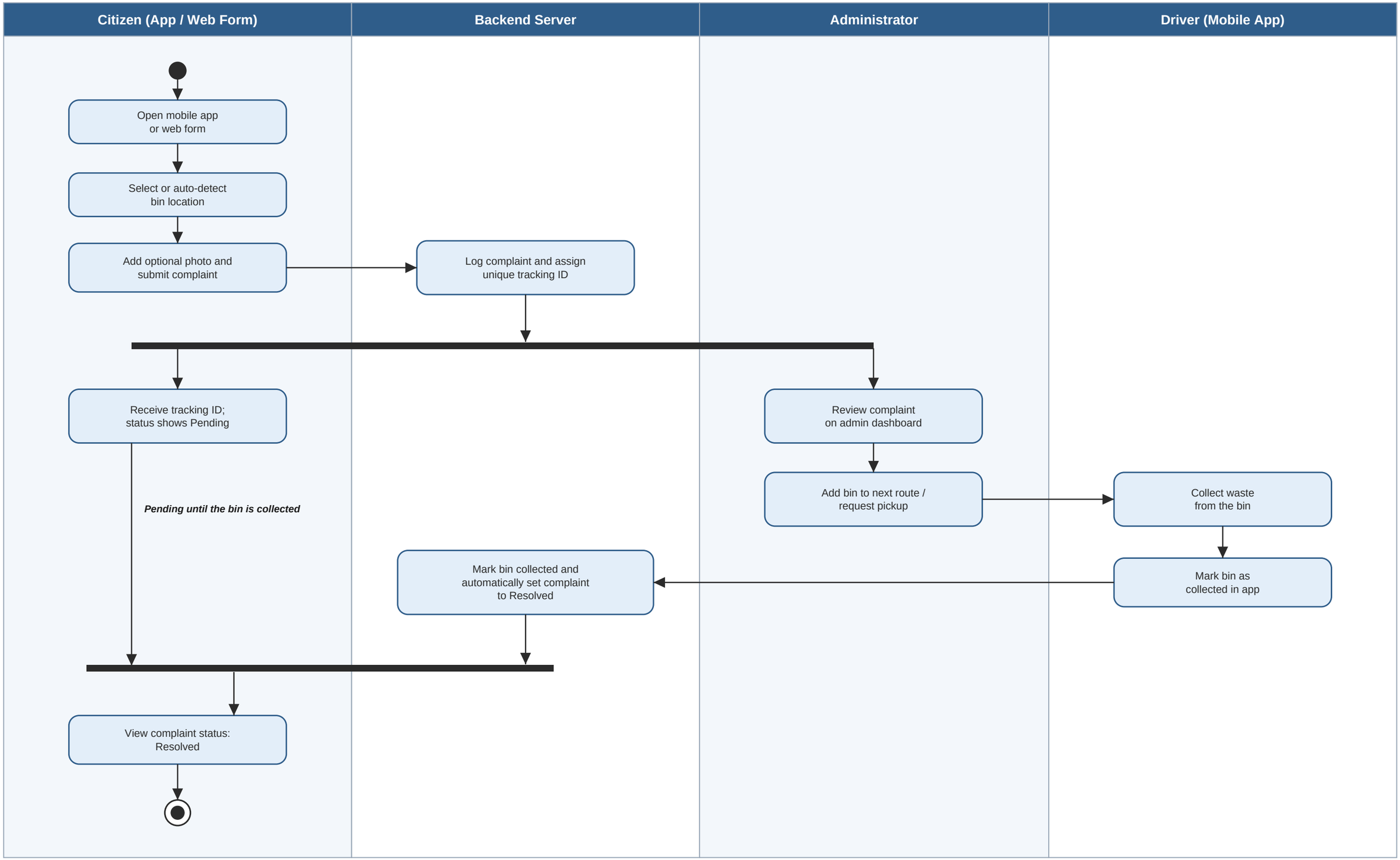

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